INCH-POUND

MIL-DTL-82962D(OS)
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July 19,2001

DETAIL SPECIFICATION

CARTRIDGES, IMPULSE, CCU-136/A, CCU-138/A, and CCU-136A/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document gives the requirements for the manufacture, assembly, and packaging of the CCU-136/A, CCU-138/A, and CCU-136A/A impulse cartridges and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or documents recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents referenced in sections 3 and 4 of this specification, whether or not those documents are listed in this section.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Department of Defense

MIL-I-23659 General Design Requirement for Electrical Initiators

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

Department of Defense

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-286	Propellants, Solid: Sampling, Examination, and Testing
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
DOD-STD-2101	Classification Of Characteristics

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Air Systems Command

842AS205	Cartridge, Impulse, CCU-136/A and CCU-138/A
842AS207	Primed Subassembly
842AS110	Output Charge, Smokeless powder
842AS210	Fused Subassembly
842AS213	Low Flash Primer Charge
842AS217	CCU-136A/A Impulse Cartridge Assembly
842AS219	Primed Assembly
842AS220	Filter Assembly
842AS221	Fused Body Assembly
842AS241	CCU-136A/A, CCU-136/A, CCU-138/A Impulse Cartridge Vibration Fixture
DL2518449	Test Set (for Mk 131 Mod 0 Cartridge)

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head Division, Documentation Branch (Code 4230), Indian Head, MD 20640-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American National Standards Institute (ANSI)/American Society For Quality Control (ASQC)

ANSI/ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(DoD activities may obtain copies from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. The private sector and other Government agencies may purchase copies from the American Society for Quality Control, P. O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

American Society For Testing And Materials (ASTM)

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ASTM E 1742 Standard Practice for Radiographic Examination
(Application for copies should be addressed to the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

American National Standards Institute

ANSI Z1.1 Guide for Quality Control Charts
ANSI Z1.2 Control Chart Method of Analyzing Data
ANSI Z1.3 Control Chart Method of Controlling Quality During Production

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.2.

3.2 General. The CCU-136/A and CCU-138/A impulse cartridges shall be manufactured in accordance with DL842AS205 and all documents and drawings listed thereon. Drawings listed in this specification are included as part of the data package defined by DL842AS205. The CCU-136A/A impulse cartridges shall be manufactured in accordance with DL842AS217 and all documents and drawings listed thereon. Drawings listed in this specification are included as part of the data package defined by DL842AS217.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidders selections shall be clearly stated in the proposal.

3.4 Primary components. For the purposes of this specification, the low flash primer charge (Drawing 842AS213) and output charge (Drawing 842AS110) are defined as primary components (see 4.3.3).

3.4.1 Igniter composition and loading. Composition shall be determined by weight rather than by volume measurement. In addition to supplier certification, new lots of material shall be independently certified to ensure accurate composition.

3.4.2 Propellant charges.

3.4.2.1 Quantity. Charges shall be determined by weight rather than by volume measurements.

3.4.2.2 Calorimetry. Using a differential scanning calorimeter at a heating rate of 10°C per minute between 100 to 350°C, the exothermic peak shall occur within 10C of baseline material known to be good (approximately 206°C).

3.4.2.3 Moisture and volatiles. Moisture and volatiles content shall be 1.0 percent maximum when tested in accordance with MIL-STD-286, Method 101.1.2.

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3.4.2.4 Quality characteristics. Main charge propellant weight shall be a quality characteristic for statistical process control. Sampling for control charting shall be conducted by the contractor in accordance with established procedures.

3.5 Subassembly requirements.

3.5.1 Insulation resistance. The insulation resistance of each fused subassembly, Drawing 842AS210 or 842AS221, shall not be less than 50 megohms at a relative humidity of less than 80 percent when measured in accordance with 4.6.1.

3.5.2 Glass-to-metal seal. The fused subassembly, Drawing 842AS210 or842AS221, shall pass a dead load, hydrostatic, and helium leak test when tested as specified in 4.6.2.

3.5.3 Capacitance. The capacitor-body assembly, drawing 842AS220, shall be subjected to capacitance measurement of 10 KHz minimum in accordance with method 305 of MIL-STD-202. The capacitance shall be 88 to 112 nano-farads when measured from pin to pin. Reject any parts not meeting this requirement.

3.6 Thermal transient test. Each primed subassembly, Drawing 842AS207 or 842AS219, shall be subjected to a thermal transient test to verify proper bridgewire and ignition mix interface. A thermal response greater than 200 microvolts shall indicate a defective subassembly (see 4.6.3).

3.7 Cartridge inspection.

3.7.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.6.4.

3.7.2 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically as specified in 4.6.5, shall not be less than 0.85 ohm and not be greater than 1.15 ohm.

3.7.3 Leakage. The cartridges shall not exhibit a leak rate in excess of 10^{-5} cc/sec of air when tested in accordance with 4.6.6.

3.7.4 X-ray. Cartridges shall be free of imperfections when examined in accordance with 4.6.7.

3.7.5 Power current. The cartridge shall not fire when subjected to one watt of power for 5 minutes when tested in accordance with 4.6.8. The power current test is considered a potential destructive test and the test shall only be conducted by the designated test activity on the first article and production lot samples. The test shall not be conducted on units intended for delivery.

3.8 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.6.10.1, shall meet the following requirements when subjected to the test prescribed in 4.6.10.1. Any cartridge failing to meet one or more requirements shall be classified as a defective unit and the lot represented rejected.

3.8.1 Velocity. Each cartridge shall be capable of ejecting a 0.46 ± 0.01 pound slug. When fired in the test device of 4.6.10.1 (see 6.5.1), the velocity for the CCU-136/A and CCU-136A/A shall be not less than 125 feet per second nor greater than 170 feet per second; the velocity for the CCU-138/A cartridge shall be not less than 90 fps nor greater than 140 fps.

3.8.2 Misfire. There shall be no misfires.

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3.8.3 Ignition delay. The ignition delay shall not be more than 15 milliseconds when tested in accordance with 4.6.10.3.

3.8.4 Post-fire. The post-fire resistance shall be greater then 500 ohms in accordance with 4.6.10.4.

3.8.5 Quality characteristics. Ballistic velocity shall be a quality characteristic for statistical process control. Sampling for control charting shall be conducted by the contractor in accordance with established procedures.

3.9 Statistical process control. Statistical process control in accordance with ANSI Z1.1, Z1.2, and Z1.3 shall be required for quality characteristics cited in this specification (main charge, ballistic velocity). In addition to other requirements for this specification, only ballistic samples from lots that can demonstrate a state of statistical control shall be submitted for lot acceptance ballistic testing. Samples from lots that cannot demonstrate statistical control shall not be submitted for lot acceptance ballistic tests.

3.10 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification and associated drawings. Particular attention should be directed to dimensions, finishes, coatings and glass-to-metal seals.

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.3.2)
b. Quality conformance inspection (see 4.5)

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of ASQC Z1.4. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with ASQC Z1.4.

4.3.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 82 cartridges shall be prepared. This sample, manufactured using the same methods and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract. Seventy-two of these cartridges shall be expended in the tests listed below and ten shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of 3.4 through 3.7.5 of this specification. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. First article sample acceptance shall consist of the following tests as prescribed in MIL-I-23659: electrical characteristics (18); temperature-shock/humidity/altitude (18); vibration (12) vibration test using vibration fixture P/N 842AS241; 20 "g" shock (12); 6-foot drop (6); 40-foot drop (6). The number in parentheses denotes the number of cartridges allocated for test. No cartridge shall produce results outside the limits set forth in 3.7 when fired in the test fixture of 4.6.8.1. Any quantity production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. The first article cartridges will not be applied as a part of the quantity specified for delivery by the contract. The units

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subjected to 40-foot drop shall not explode or lose propellant and shall be safe to handle for disposal. They will not be ballistically tested.

4.3.3 Production lot. The suggested production lot size is 19,280 (40 boxes) cartridges. Lot numbers shall be in accordance with MIL-STD-1168. Cartridges required for all test purposes are randomly selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting agency. The designated test activity shall perform the cartridge tests and examinations listed in table I to determine lot acceptance.

4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL842AS205 or DL842AS217 and the tests and examinations of table I.

4.5.1 Packing, packaging and marking. The inspector shall ascertain that the packing and packaging of the cartridges and the container markings conform to Section 5 of this specification.

4.6 Tests.

4.6.1 Insulation resistance. Prior to assembly of the bridgewire and filter, the insulation resistance shall be checked with an applied voltage of 500 ± 25 volts dc. Voltage to be applied between pins and between shorted pins and case. Units which fall to meet the requirements of 3.5 shall be rejected.

4.6.2 Glass-to-metal seal. The glass-to-metal seal shall withstand 320 pounds minimum dead load applied over a 0.160 ± 0.005 inch diameter at center without cracking when applied as shown on Drawing 842AS210 or 842AS221. The seal shall also withstand a 5000 ± 500 psi hydrostatic pressure without cracking or leaking. Following both these tests (dead load and hydrostatic), the seal shall not exceed 1 x 10^{-6} cc/sec air when tested at a pressure differential of 1.0 ± 0.1 atmosphere.

4.6.3 Thermal transient test. Each Primed Subassembly (Drawing 842AS207 or 842AS219) shall be tested using an 80 ± 10 millisecond step input set for 160 milliamps across a 1-ohm resistor. The differential thermal response to the step input shall be captured.

4.6.4 Visual inspection. Examine each cartridge produced and reject for bent connector pins, perforated or improperly crimped closure discs, incorrect or illegible markings or other visible defects which might adversely affect cartridge installation or ballistic properties.

4.6.5 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.7.2. shall be ascertained by means of test circuit which limits the bridge current to 25 milliamperes, maximum. Any cartridges not meeting the requirement shall be rejected. The test circuit shall be connected to the cartridge through a fully assembled electrical connector or equal. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is met. Units which fail to meet the requirements of 3.7.2 or fail to

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assemble completely, by hand, with the specified connector shall be rejected. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

TABLE I. Quality conformance inspections.

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Primary component	Single lot verification	3.4	4.3.3	(M101)
Fused subassembly	Insulation resistance	3.5.1	4.6.1	(M102)
Fused subassembly	Glass-to-metal seal	3.5.2	4.6.2	(M103)
Primed subassembly	Thermal transient test	3.6	4.6.3	(C1)
Ignition mix	Compounding and loading	3.4.1	C	(M104)
Propellant charge	Type and weight verification	3.4.2	C	(M105)
Cartridge	Visual inspection	3.7.1	4.6.4	(M106)
Cartridge	Bridge circuit resistance	3.7.2	4.6.5	(M107)
Cartridge	Leakage	3.7.3	4.6.6	(M108)
Cartridge	X-ray	3.7.4	4.6.7	(M109)
Cartridge	Power current	3.7.5	4.6.8	(M110)
Cartridge	Ballistic requirements	3.8	4.6.10	(C2)
Cartridge	Packing, packaging, and marking	Section 5	4.5.1	(M111)

4.6.6 Leakage. Leak test each cartridge in the first article sample and ballistic sample in a dry gas leak tester in accordance with paragraph 4.6.6.1. CCU-136/A andCCU-138/A impulse cartridges that exhibit a leak rate in excess of 10^{-5} cc/sec of air at a pressure differential of one atmosphere are considered defective. Retest CCU-136A/A impulse cartridges that exhibit a leak in excess of 10^{-5} cc/sec of air at a pressure differential of one atmosphere in accordance with paragraph 4.6.6.2. If one or more cartridges of the ballistic sample fail the leak test when tested in accordance with paragraph 4.6.6.1 or 4.6.6.2, a leak test of each cartridge within the lot shall be required. If a 100 percent leak test of the lot is required, it shall be at the expense of the contractor. Cartridges exhibiting a leak rate in excess of that specified above shall be rejected.

4.6.6.1 Cartridge leak procedure.

a. Place cartridges in a pressure vessel. [NOTE: Leak testing of cartridges must take place within 15 + 1 minute from the time you remove the cartridge from the pressure vessel. Therefore, process only as many cartridges as you can leak test within 15 + 1 minutes of helium bombing. Re-pressurize any cartridge not tested within 15 + 1 minutes.]

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b. Purge the pressure vessel or evacuate the pressure vessel to a pressure of less than 4 inches of mercury (2 psia) and close vacuum valve.

c. Pressurize the vessel with helium to 30 ± 2 psia for 60 +1/-0 minutes.

d. Remove the test cartridges from the pressure vessel and record the time.

e. Flush exterior of the test cartridge with low pressure air to remove possible trapped helium,

f. Perform leak test on cartridges using helium mass spectrometer leak detector.

4.6.6.2 Output closure leak test procedure.

a. Place cartridges in a pressure vessel. [NOTE: Leak testing of cartridges must take place within 15 + 1 minute from the time you remove the cartridge from the pressure vessel. Therefore, process only as many cartridges as you can leak test within 15 + 1 minutes of helium bombing. Re-pressurize any cartridge not tested within 15 + 1 minutes.]

b. Purge the pressure vessel or evacuate the vessel to a pressure of less than 4 inches of mercury (2 psia) and close vacuum valve.

c. Pressurize the vessel with helium to 30 ± 2 psia for 60 +1/-0 minutes.

d. Remove the test cartridges from the pressure vessel and record the time.

e. Flush exterior of the test cartridge with low pressure air to remove possible trapped helium,

f. Perform leak test on cartridges using helium mass spectrometer leak detector and fixture P/N 842AS239.

4.6.7 Radiographic examination. All cartridges in the first article or production lot test samples shall be radiographically examined in accordance with ASTM E 1742; any observable imperfections in assembly shall be cause for cartridge rejection. The cartridge shall be positioned on their sides for the most revealing exposure with the plane of the pins perpendicular to the X-ray source. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on tray boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. If any assembly or loading defects are noted then the contractor must 100 percent X-ray the represented lot per 4.6.6 and submit results to the cartridge design activity for review. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings. When screening of the production lot by X-ray review is required, it shall be at the expense of the contractor.

4.6.8 Power-current test. Each cartridge in the first article and production test sample shall have a direct current of not less than one ampere supplying a minimum of one watt applied to the bridge circuit for at least 5 minutes. The test current shall be regulated throughout the period of application to within 2 percent. If a rectified current is used, the ripple content shall not exceed 5 percent rms of the test current.

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Test temperature is ambient and the first article sample or production lot is rejected if a cartridge fires within 5 minutes.

4.6.10 Ballistic test.

4.6.10.1 Test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with ASQC Z1.4. The suggested sample size for a proven manufacturer is 190 cartridges for ballistic test and 3 for investigation purpose. The randomly selected cartridges shall be conditioned at 70° ± 5°F, –65° ± 5°F, or 160° ± 5°F for 4 hours minimum and fired in a test fixture, Drawing 2518449. Firing current shall be 5.0 ± 0.1 ampere. Instrumentation shall be provided to measure the slug velocity and the time interval between the application of firing current and the initial movement of the slug from the tube muzzle. Velocity shall be determined from the flight time to cover the initial 36 ± 0.125 inches of slug travel.

4.6.10.2 Acceptance criteria. The total number of allowed defective units in a first article sample or lot acceptance sample as a result of tests and inspections of 3.7 through 3.9 shall be zero. Any defects will reject the lot represented. Screening of defects as a corrective action is at contractor's expense and the method of screening shall be subject to approval by the contracting officer.

4.6.10.3 Ignition delay. When the cartridges are fired in the test fixture, Drawing 2518449, the time from application of current to the first indication of slug movement shall not exceed 15 milliseconds at any of the specified conditioning temperatures of –65° to 160°F.

4.6.10.4 Post-fire. The resistance between the pins of the cartridges shall be measured with a multimeter.

4.6.10.5 Retest. There shall be no retests.

4.6.10.6 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and the complete test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The CCU-136/A, CCU-138/A, and CCU-136A/A impulse cartridges are designed to provide the power to eject expendable countermeasures from aircraft. The impulse cartridges are designed for use on military aircraft only, and thus have no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. Designation of impulse cartridge (CCU-136/A, CCU-136A/A or CCU-138/A).

d. Whether first article inspection is required and, if so, specify the test activity.

e. Production lot size and test activity (see 4.3.3).

f. Items of data required for each first article and production lot (see 6.3)

g. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore," OP 5 are applicable.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order.

6.3.1 Standard deliverables. In addition to the delivery of cartridges, the contract may require delivery of the following:

a. Radiographic plates
b. Radiographic review report
c. Radiographic sample plates and techniques
d. Requests for waiver/deviation and engineering change proposal (ECP)
e. Progress and management report
f. Test and inspection report if contractor tested
g. Ammunition data cards
h. Performance oriented packaging (POP) test report
i. Process control charts.

6.4 Additional data.

6.4.1 Closed bomb performance. The CCU-136/A and CCU-136A/A provide approximately 614 psi at ambient temperatures in a 43.5 cc Closed bomb. Time to peak pressure is approximately 4.4milliseconds.
6.4.2 All fire performance. The CCU-136/A impulse cartridge has a minimum all-fire level of 2.63 amps (Bruceton Test Method) with 5-minute current pulse duration. Another test (Langlie) determined the minimum all-fire level to ignite the cartridge within 15 ms to be 2.89 amps. Testing with the CCU-136A/A determined a minimum all-fire current of 2.21 amps with 50 ms pulses (Bruceton).

6.4.3 No-fire performance. The cartridges have a maximum no-fire of 1.40amps (Bruceton) with 5-minute current pulse duration. Testing with the CCU-136A/A determined a maximum no-fire current of 1.88 amps with 50 ms pulses (Bruceton).

6.4.4 Qualification data. Indian Head Test Report IHTR 1966 documents the qualification testing for the CCU-136/A and the CCU-138/A impulse cartridges. Indian Head Test Report IHTR2204 documents the qualification testing of the CCU-136A/A impulse cartridge.

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6.5 Definition.

6.5.1 Velocity. Velocity is defined as the distance from the tube muzzle to a point 36 ± 0.125 inch beyond the tube muzzle divided by the slug transit time between these two points.

6.6 Differential scanning calorimetry. ASTM E 793 may be used as guidance for conducting tests and interpreting data.

6.7 Subject term (keyword) listing.

chaff
countermeasures
decoy
flare
jammer
stores release

6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue. The following ECPs were incorporated in this revision; 025220R004R1, and 02522R027.

Preparing activity:
Navy - OS
(Project 1377-0173)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-82962D (OS)	2. DOCUMENT DATE (YYMMDD) 020820

3. DOCUMENT TITLE
CARTRIDGES, IMPULSE, CCU-136/A, CCU-138/A, and CCU-136A/A

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Engineering Documentation Branch (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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